DATA ITEM DESCRIPTION
Title: POD MASTER EVENT SPECTRUM (MES)

Number: DI-SESS-82266
AMSC Number: N10031
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20190516
Limitation: None
GIDEP Applicable: No
Project Number: SESS-2019-019

Use/relationship: The Pod Master Event Spectrum describes the analysis of the effects from repeated loads experienced on the pod's structure. The data will be used to verify the pod's ability to meet the durability, damage tolerance, and operational service life requirements specified in the contract.

This Data Item Description (DID) contains the format and content for data resulting from the work tasks described by the contract statement of work.

Requirements:

1. Reference documents. None

2. Format. Contractor format acceptable.

3. Content. The Pod MES shall contain the following:

3.1 Describe the aircraft MES that forms the basis for the pod MES. The aircraft Master Event Spectrum (MES) is a profile of specific sequences and number of aircraft occurrences that reflect the fatigue exposure of the pod over a lifetime. It includes events such as taxis, takeoffs, flight maneuvers, and landings that are specific to a particular aircraft and set of missions.

3.2 Describe pod specific modifications and additions to the aircraft MES used to create the pod MES. Define the pod coordinate system and show how the aircraft MES is translated to the pod coordinated system. Define any factors beyond simple geometric translation that define the conversion of the aircraft MES to the pod MES, e.g. structural dynamic effects between the aircraft MES reference location and the pod MES reference location. Define pod specific events that alter loads in terms of sequence, duration, and number of occurrences, such as drag increase due to power generation or external surface actuations, or any pod event that alters external loads from the basic aircraft carriage loads.

3.3 Describe the external loads time histories for each event in the pod MES. Include the reference coordinate system, six degree-of-freedom (6DOF) loads and accelerations, inertial and air loads, quasi-static shock loads, and any dynamic buffeting occurring.

3.4 Describe methods and analytical tools used to perform the analysis, including rainflow

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counting methodology and any filters used to reduce or eliminate events from the pod MES.

End of DI-SESS-82266

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